

TABLE OF CONTENTS

Table of Contents	1
1- Scan Room Interface (SRI) Board, MG2 A33 A1	2
1-1 SRI Board Removal	2
1-2 SRI Board Replacment	3
2- Dock Limit Switch Board (MG2 A31 A1)	5
3- Dock Components.....	6
3-1 Dock Assembly	7
3-1-1 Dock Assembly Removal	7
3-1-2 Dock Assembly Installation.....	11
3-2 Dock Motor Removal and Replacement.....	12
3-4 Dock Motor On Switch Removal.....	14
3-5 Dock Motor On Switch Replacement.....	15
3-6 Table Up Limit Switch Removal.....	17
3-7 Table Up Limit Switch Replacement.....	18
Revision History	20

Description - Contains replacement procedures for the Scan Room Interface (SRI) board, located in the magnet enclosure and the table dock hardware.

1- SCAN ROOM INTERFACE (SRI) BOARD, MG2 A33 A1

RF/PEN Cabinet:

For SRI voltage adjustment procedures, refer to procedure *Magnet Enclosure P.S. Module Calibration* (RC1SCA4.PDF).

RF/PEN II Cabinet:

There are no SRI adjustments for the RF/PEN II Cabinet.

1-1 SRI Board Removal

1. Lock-Out/ Tag-Out circuit breaker for **RF/Pen Cabinet** or **RF/PEN II Cabinet** at the PDU using appropriate OSHA procedure. (Refer to Safety procedure, SYSMSA2.PDF.)
2. Verify that voltage between +24V SRI and +24V SRI RTN is 0 VDC for the **RF/PEN Cabinet** or **RF/PEN II Cabinet**.
3. Remove right-front side panel of magnet enclosure to gain access to SRI.
4. Loosen retaining screws on front cover of SRI, and slide front cover open to expose SRI board.

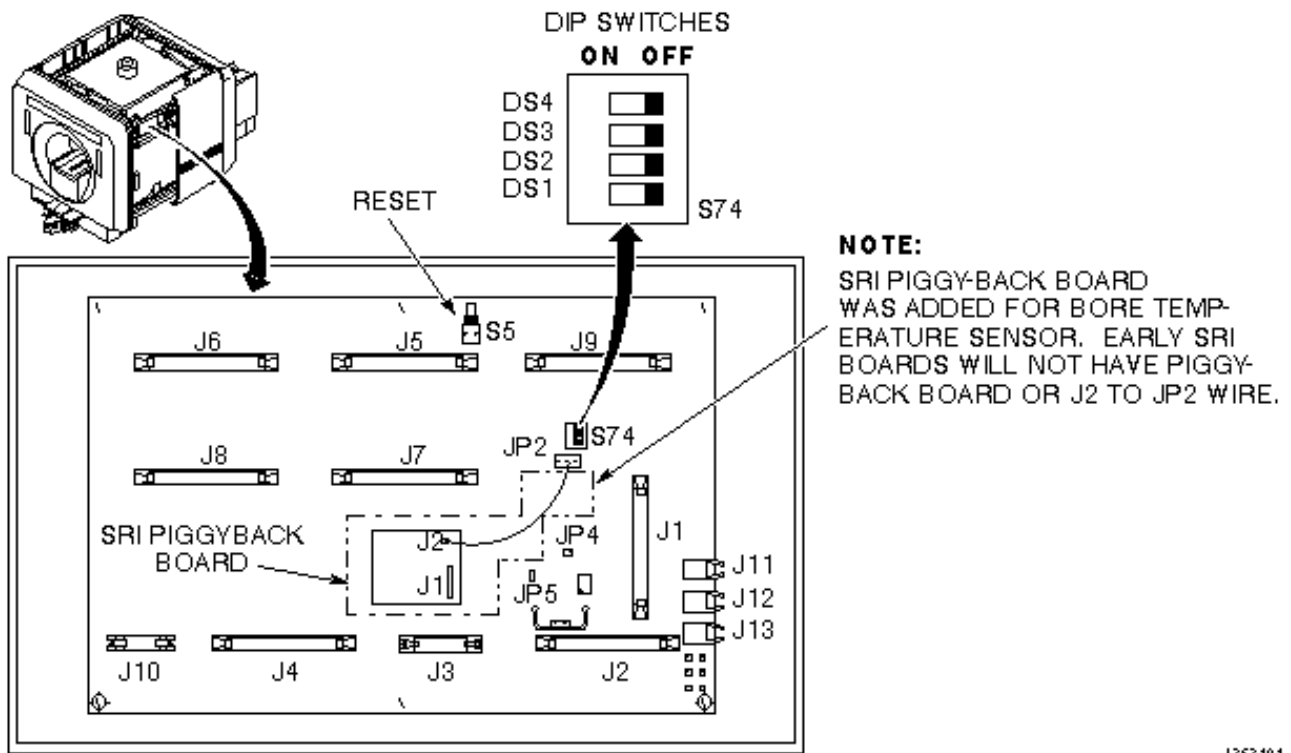
Note

If the system has already been powered up, disconnecting the SRI board will put the system into EMERGENCY STOP condition.



Possible Equipment Damage! To avoid static damage to circuit board components, wear a wrist-grounding strap when handling the SRI board.

5. Disconnect all ribbon cables from SRI board. (See Illustration 1-1.)



6. Disconnect fiber-optic cables from connectors J11, J12, and J13 on the SRI board. (See Illustration 1-1.)
7. Using a nut driver, remove the six nuts and lock washers from the board stand-off studs.
8. Slide board off the studs and remove it.

1-2 SRI Board Replacment

1. Verify proper jumper and switch settings on the replacement SRI board. (Refer to Tables 1 and 2.)

TABLE 1
SRI BOARD SWITCH INFORMATION

Switch	Schematic Location	Position	Description	Notes
S5	2-B4	Open	Board Reset	
S74-SW1	2-H9	Open	(Test Use)	
S74-SW2	2-H9	Open	(Test Use)	
S74-SW3	2-H9	Open	(Test Use)	
S74-SW4	2-H9	Open	(Test Use)	

TABLE 2
SRI BOARD JUMPER INFORMATION

Jumper	Schematic Location	Position	Description	Notes
JP2	2-F6	B	Polarity Source (SRI without piggyback board)	Selects source for LCD Polar signal. Position A selects the 8OC196 as source; position B selects off-board source.
JP2	2-C5	Default-Installed	XTAL Test	Provided for testability
JP5	2-E9	Default-Installed	Memory Select	Configures memory accessibility. Jumper not installed allows stand-alone diagnostics shell to be accessed. Jumper installed allows normal operation.

- Slide replacement SRI board onto studs.
- Replace lock washers and nuts. Tighten nuts with nut driver.
- Reconnect fiber optic cables to connectors J11, J12, and J13.
- Connect all ribbon cables to SRI board.
- Slide SRI front cover closed and tighten retaining screws.
- Replace right-front side panel of magnet enclosure.
- Remove Lock-Out / Tag-Out from circuit breaker MR1 A14 CB6 at rear of **RF/PEN Cabinet** or MR1 A20 CB4 at the rear of the **RF/PEN II Cabinet**.
- Restore power to SRI by switching on circuit breaker MR1 A14 CB6 at rear of **RF/PEN Cabinet** or MR1 A20 CB4 at the rear of the **RF/PEN II Cabinet**.
- Refer to procedure for *Enclosure Indicators & Control Checks* (ME1FCA1.PDF) and perform all tests. (Refer to Table 1 (switches), Table 2 (jumpers) above, and Table 3 (test points) and Table 4 (LEDs) for service component information.

TABLE 3
SRI BOARD TEST POINT INFORMATION

Test Point	Schematic Position	Description	Notes
TP1	2-D3	RCV Data	
TP2	2-D3	RCV Data	
TP3	2-C1		Reset
TP4	5-C9		Encoder Voltage P
TP5	5-C9		Encoder Voltage N

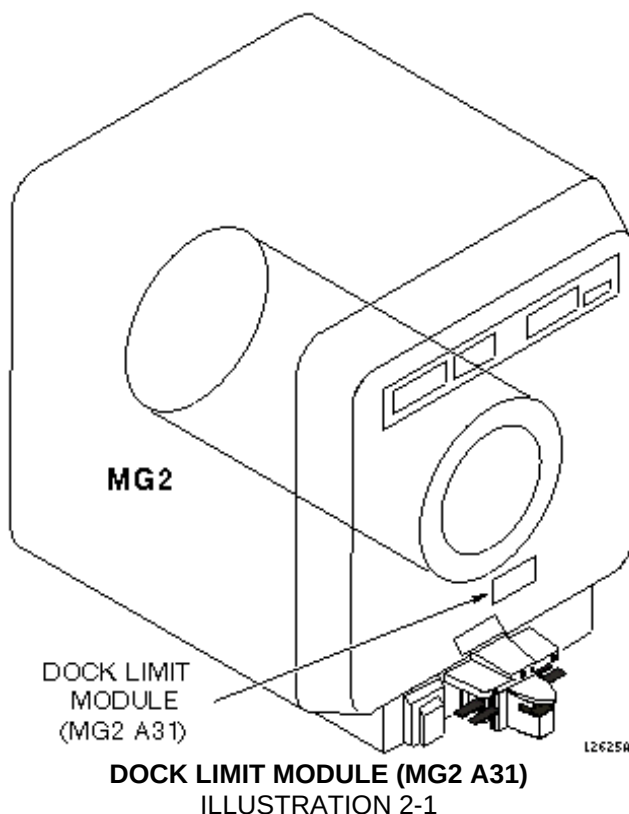
TABLE 4
SRI BOARD LED INFORMATION

LED	Schematic Position	Color	Description	Notes
DS29	2-E8	Green	Motor Disable	
DS30	2-E8	Yellow	Test	
DS31	2-F5	Green	Uplim	
DS32	2-F4	Green	Home	
DS33	2-F4	Green	EOT	
DS34	2-F4	Green	Dckstp	
DS116	2-D3	Green	Enc-B	
DS117	2-D3	Green	Enc-A	

2- DOCK LIMIT SWITCH BOARD (MG2 A31 A1)

The Dock Limit Switch Board (MG2 A31 A1) is located in front of the magnet enclosure, inside the Dock Limit Module.

1. Lock-Out / Tag-Out circuit breaker MR1 A14 CB6 at rear of **RF/Pen Cabinet** or circuit breaker MR1 A20 CB4 at rear of **RF/Pen II Cabinet**. (Refer to Safety procedure, SYSMSA2.PDF.)
2. Verify that voltage between +38.5V Dock and Dock/Light RTN is 0 VDC for the **RF/PEN Cabinet** or **RF/PEN II Cabinet**.
3. Undock the patient transport and slide magnet enclosure front cover forward to access the Dock Limit Module (MG2 A31). (See Illustration 2-1.)



4. Remove Dock Limit Module cover to access Dock Limit Switch board mounted inside.
5. Disconnect J1, J2, and J3 from board.
6. Remove mounting screws and replace failed Dock Limit Switch board.
7. Connect J1, J2, and J3 to new board.
8. Install Dock Limit Module cover.
9. Close and secure magnet enclosure front cover.
10. Remove Lock-Out / Tag-Out from circuit breaker MR1 A14 CB6 at the rear of the **RF/Pen Cabinet** or MR1 A20 CB4 at the rear of the **RF/Pen II Cabinet**.
11. Switch on circuit breaker MR1 A14 CB6 at the rear of the **RF/Pen Cabinet** or MR1 A20 CB4 at the rear of the **RF/Pen II Cabinet**.

3- DOCK COMPONENTS



WARNING!

STRONG MAGNETIC FIELD! WHEN SERVICING ANY MAGNETIC EQUIPMENT, IT IS CRITICALLY IMPORTANT THAT THE SERVICE ENGINEER CONSCIOUSLY PLAN THE PATH TO BE TAKEN WHEN MOVING HIGHLY FERROUS DEVICES IN THE MAGNET ENVIRONMENT. THE PATH SHOULD BE AS FAR FROM THE MAGNET AS PRACTICAL.

SAFETY REQUIREMENTS:

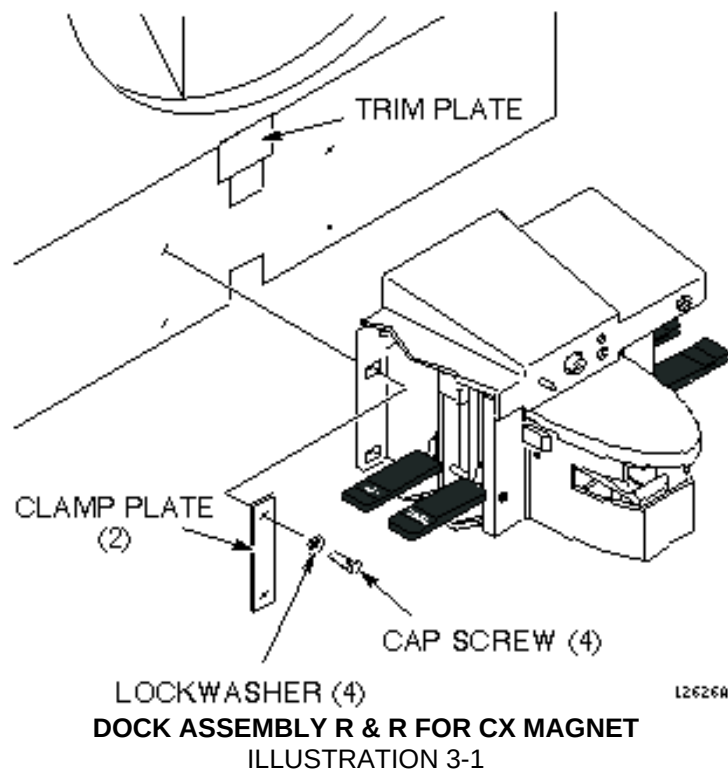
- **THE SERVICE PATH MUST ALLOW FOR THE DOCK TO BE MOVED ALONG THE GROUND A DISTANCE EQUAL TO THAT OF THE DISTANCE BETWEEN THE MAGNET AND THE FOOT-END OF THE PATIENT TABLE PRIOR TO BEING LIFTED VERTICALLY.**
- **TWO (2) MR SAFETY TRAINED PERSONNEL MUST BE PRESENT AT ALL TIMES WHEN SERVICING HIGHLY FERROUS DEVICES IN THE AREAS OF MAGNETIC FIELDS.**

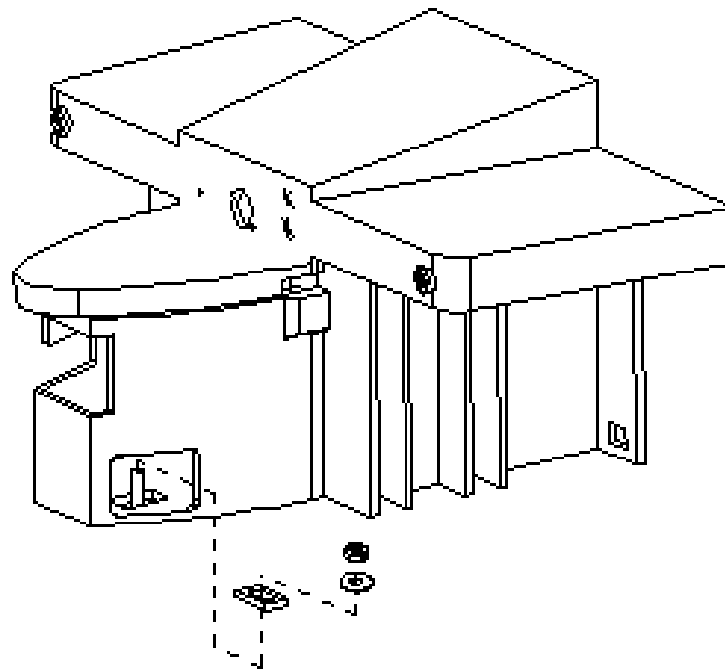
WHEN PLANNING A SERVICE PATH, IT IS CRITICAL THAT THE PATH BE CLEAR AND SUFFICIENTLY WIDE. ENSURE THAT THERE ARE NO TRIP HAZARDS, OBSTACLES, CLUTTER, SLIPPERY SURFACES OR OTHER ITEMS EVEN PARTIALLY RESTRICTING THE PATH. IF THERE ARE PORTABLE OBSTACLES IN A PATH, REMOVE THEM FROM THE AREA AND REPLACE THEM AFTER THE SERVICE ACTION IS COMPLETED. IT IS REQUIRED TO WALK THE PATH PRIOR TO BEGINNING SERVICE TO ENSURE THAT THERE IS SUFFICIENT SPACE THROUGH WHICH TO PASS FOR YOURSELF AND THE OBJECT BEING SERVICED.

3-1 Dock Assembly

3-1-1 Dock Assembly Removal

1. Lock-Out / Tag-Out circuit breaker MR1 A14 CB6 at rear of **RF/Pen Cabinet** or circuit breaker MR1 A20 CB4 at rear of **RF/Pen II Cabinet**. (Refer to Safety procedure, SYSMSA2.PDF.)
2. **RF/PEN Cabinet:** Verify that voltage between +38.5V Dock and Dock/Light RTN is 0 VDC for **RF/PEN Cabinet** or **RF/PEN II Cabinet**.
3. Undock the patient transport.
4. With two people applying downward pressure to the top of the dock assembly, unfasten the dock assembly from dock support by removing cap screws, lock washers, and clamping plates. (See Illustrations 3-1 and 3-2.)





CX DOCK FLOOR MOUNT
ILLUSTRATION 3-2

WARNING!

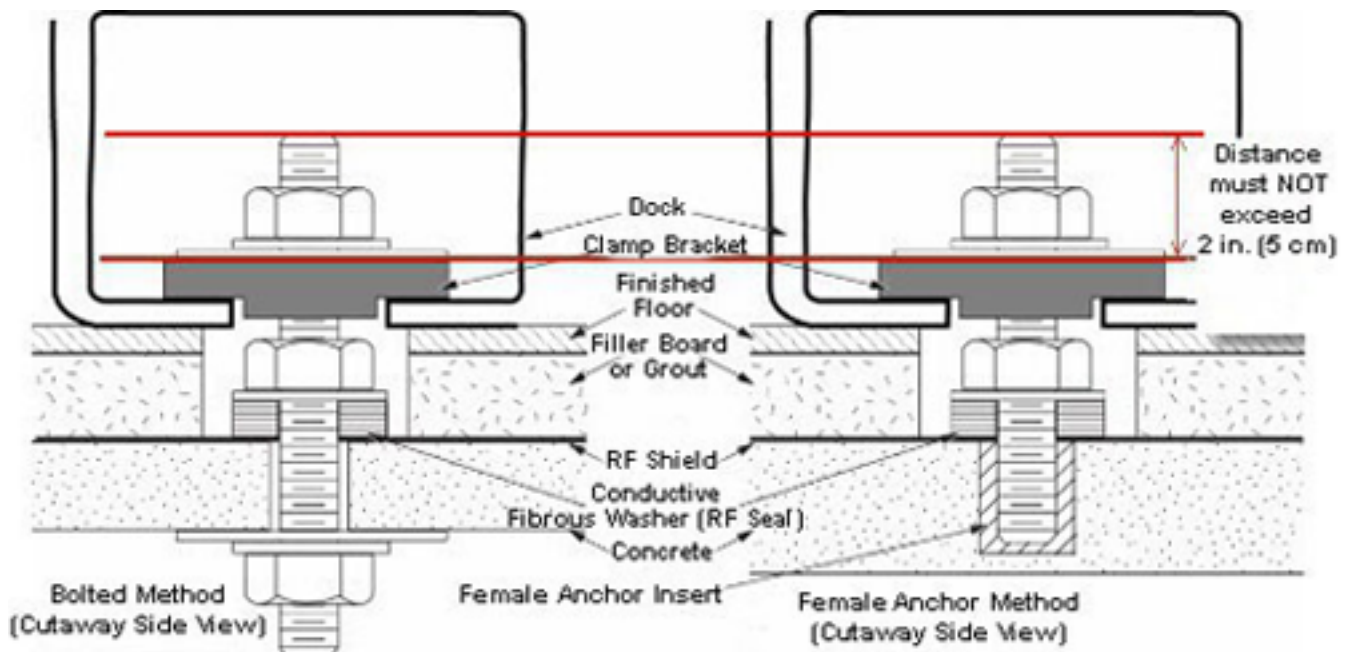
POSSIBLE PERSONAL INJURY! THE DOCK CONTAINS FERROUS MATERIALS. FROM THIS POINT OF THE PROCEDURE FORWARD, DOWNWARD PRESSURE MUST BE CONTINUOUSLY APPLIED TO THE DOCK SO AS TO PREVENT IT FROM BEING PULLED INTO THE MAGNET BORE.

5. With two people applying pressure to the top of the dock, slide dock assembly away from magnet enclosure to gain access to cable connections.
6. Remove small trim plate on front enclosure skirt, just above cutout to access dock motor power cable and UP LIMIT switch cable connections. (See Illustration 3-1.)
7. Disconnect Cable MG2 A29-P1 by removing two screws on each connector.
8. Disconnect cable from J2 on Dock Limit Module MG2 A31.



FERROUS MATERIAL HAZARD! THE DOCK ASSEMBLY CONTAINS A MOTOR MADE WITH FERROUS COMPONENTS. HOLDING THIS ASSEMBLY TOO CLOSE TO THE MAGNET BORE WILL CAUSE IT TO BE FORCIBLY ATTRACTED TO THE MAGNET. TO PREVENT POSSIBLE BODILY INJURY OR DAMAGE TO EQUIPMENT, PLACE DOCK ASSEMBLY ON FLOOR AND SLIDE IT OUT OF SCAN ROOM.

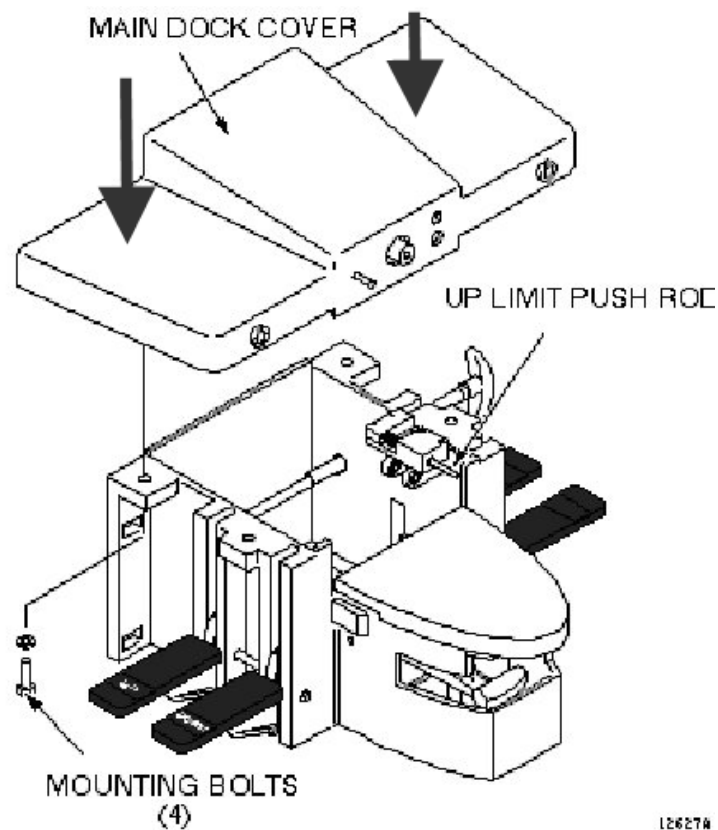
9. Clear the anchor stud out of the path of the dock assembly. This can be done in two ways:
 - a. If the site has followed the pre-installation manual, then it should be possible to remove the top portion of the stud so that the dock may remain flush with floor during removal. This is the preferred method.
 - b. If the site has an anchor stud that protrudes vertically from the floor and cannot be removed, a nonconformance **must** be raised with the customer and the anchor must be fixed to comply with the pre-installation manual prior to any future servicing of the dock. To complete the task this time, follow these steps:
 1. Measure the height of the anchor stud as indicated in the illustration below. If the height exceeds 2 inches (5 cm), the magnet must be ramped down **prior** to removing or installing the Dock as there is significant risk of the magnet attracting the Dock if it needs to be raised more than 2 inches (5 cm).



MEASURING HEIGHT OF ANCHOR STUD
ILLUSTRATION 3-3

2. Use two people to pull the Dock away from the magnet until the stud in the front is hitting the inner frame.

3. Use two people to lift the dock **just enough** to clear the top of the stud in the floor. Pull the dock away from the magnet as far as possible.
 4. Replace the dock on the floor immediately to the side of the anchor stud.
- c. With two field engineers, one holding each side of the dock (shown assembled in Illustration 3-2), slowly and deliberately slide the dock away from the magnet, keeping downward pressure on the top of the dock assembly shown with arrows in Illustration 3-4. Move it to a distance equivalent to where the foot of the patient table would normally be located.



APPLYING DOWNWARD PRESSURE AT TOP OF DOCK
ILLUSTRATION 3-4

- d. When the distance between the dock and the magnet is at least equivalent to the distance between the magnet and where the foot of the patient table would normally be located, both field engineers may slowly lift the dock from the floor and exit the room.

3-1-2 Dock Assembly Installation

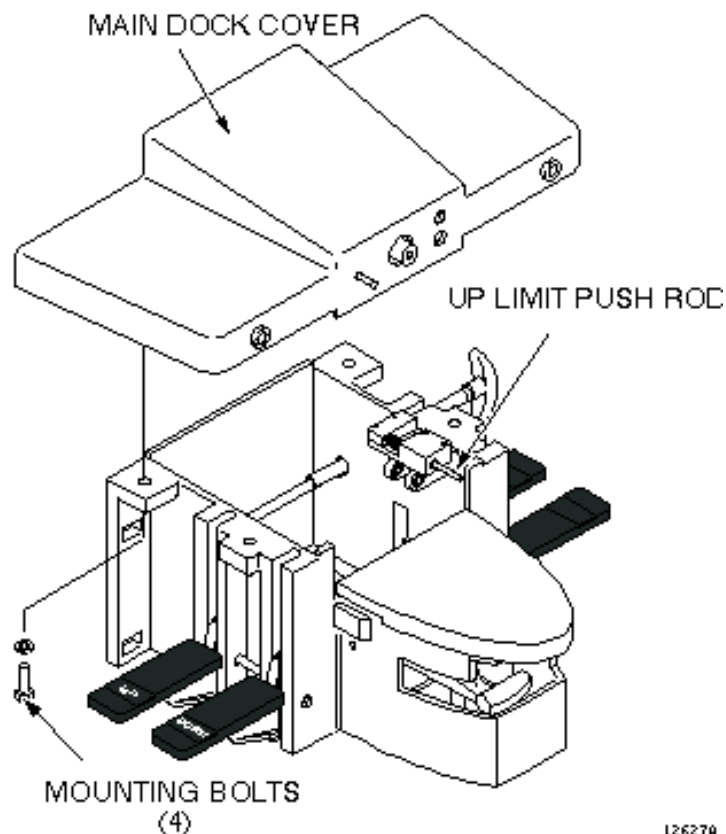


POSSIBLE PERSONAL INJURY! THIS PROCEDURE REQUIRES MOVEMENT OF FERROUS MATERIAL IN CLOSE PROXIMITY TO THE MAGNET. THEREFORE, TWO FIELD ENGINEERS ARE REQUIRED AT ALL TIMES.

1. With two field engineers, one holding each side of the Dock, enter the Magnet Room and place the dock on the floor where the foot of the patient table would normally be located.
2. With two field engineers, one applying vertical downward pressure to each side of the Dock, slide the dock toward the magnet front.
3. While maintaining pressure to the top of the dock, position the dock against the base of the magnet to the left of the floor stud at the front of the dock
4. Lift the Dock just enough to clear the top of the stud in the floor.
5. Fasten cap screws, lock washers and clamping plates.
6. Reconnect Cables MG2 A29-P1 and MG2 A31-J2.
7. Reattach the small trim plate on front enclosure skirt, just above cut-out.
8. Reinstall dock assembly to magnet enclosure with attaching hardware.
7. Remove Lock-Out / Tag-Out from circuit breaker MR1 A14 CB6 at the rear of the **RF/Pen Cabinet** or MR1 A20 CB4 at the rear of the **RF/Pen II Cabinet**.
8. Switch on circuit breaker MR1 A14 CB6 at the rear of the **RF/Pen Cabinet** or MR1 A20 CB4 at the rear of the **RF/Pen II Cabinet**.
9. Dock patient transport.
10. Refer to procedure for *Lightweight Patient Transport, Section 4: Dock Centering* (PT1SCA2.PDF) to determine dock centering requirements.

3-2 Dock Motor Removal and Replacement

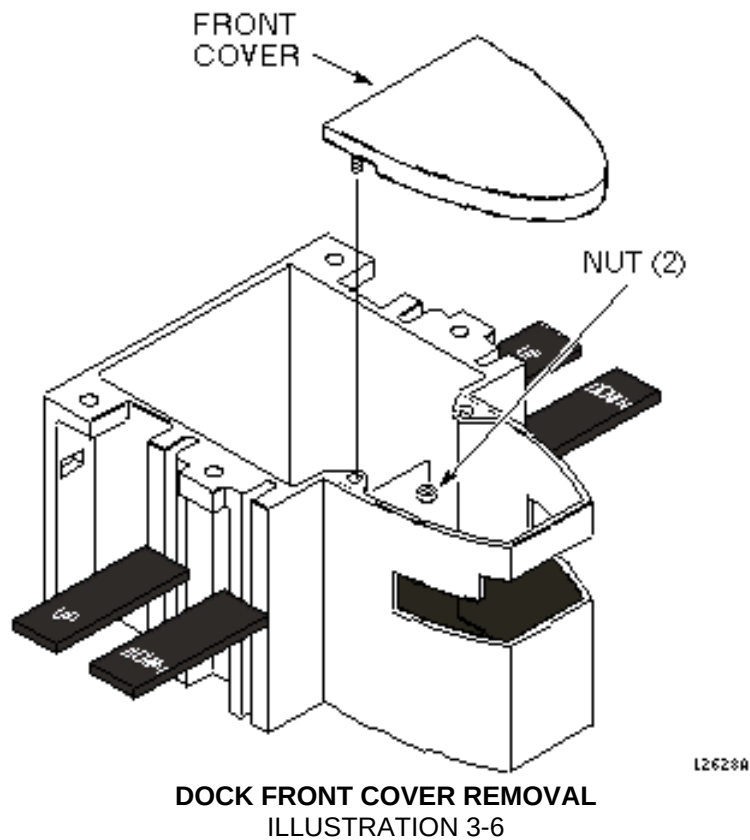
1. Refer to *Section 3-1-1, Dock Removal* instructions.
2. Remove four mounting bolts from main dock cover. (See Illustration 3-5.)



MAIN DOCK COVER REMOVAL
ILLUSTRATION 3-5

3. Slide the main dock cover forward until Up Limit Switch pushrod just clears main dock cover holes, then lift cover up. Unplug the dock motor power lead plug before totally removing the dock cover.

4. Remove front cover by removing two mounting nuts. (See Illustration 3-6.)



5. Disconnect power lead to dock motor.
7. Remove four bolts from yoke securing dock motor assembly to main dock cover. (See Illustration 3-5.)
8. Remove dock motor assembly from main dock cover.
9. Apply Loctite® 271 to dock motor assembly mounting bolts.
10. Install replacement dock motor assembly on main dock cover by securing four mounting bolts through yoke assembly. (See Illustration 3-5.)
11. Replace front dock cover by securing two nuts. (See Illustration 3-6.)
12. Connect power lead to dock motor.
13. Carefully position main dock cover so that Up Limit Switch pushrod is aligned with upper hole in dock cover.
14. Replace main dock cover by securing four mounting bolts. (See Illustration 3-5.)

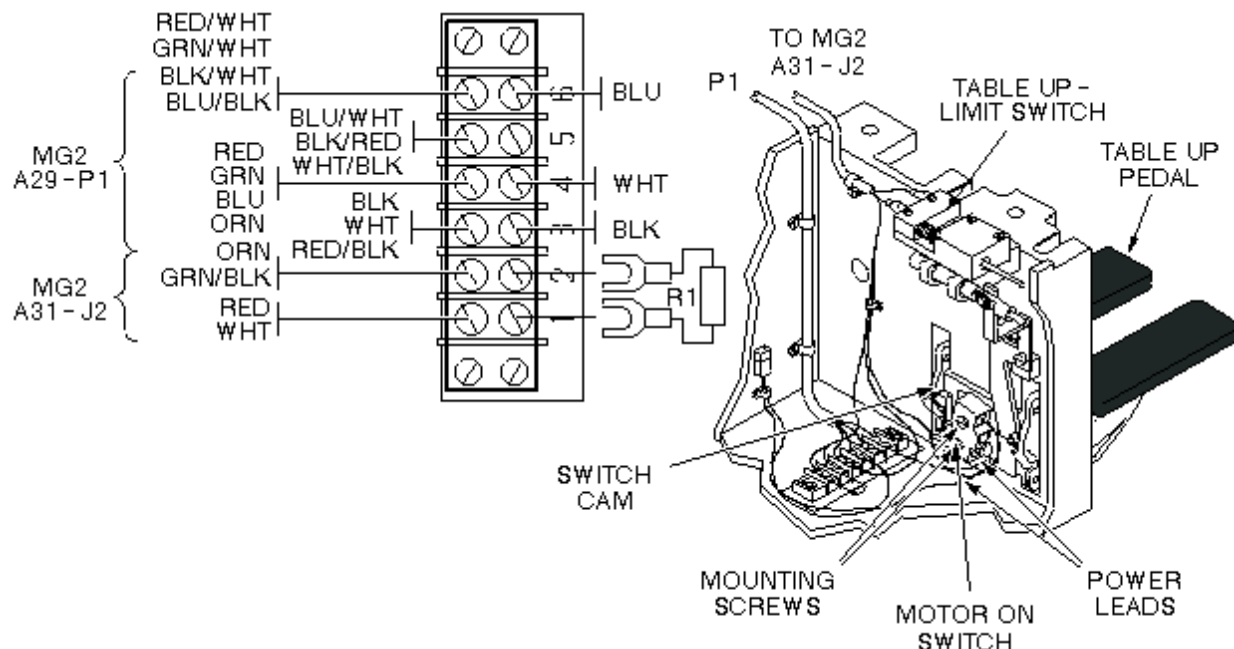


FERROUS MATERIAL HAZARD! THE DOCK ASSEMBLY CONTAINS A MOTOR MADE WITH FERROUS COMPONENTS. HOLDING THIS ASSEMBLY TOO CLOSE TO THE MAGNET BORE WILL CAUSE IT TO BE FORCIBLY ATTRACTED TO THE MAGNET. TO PREVENT POSSIBLE BODILY INJURY OR DAMAGE TO EQUIPMENT, PLACE DOCK ASSEMBLY ON FLOOR AND SLIDE IT OUT OF SCAN ROOM.

15. Refer to *Section 3-1-2* for Dock installation instructions.
16. Check operation of dock motor by performing procedure for Enclosure Covers, Dock Adjustments: *Section 1, Dock Motor Speed Adjustment* and *Section 4, Dock Motor Gear Assembly Adjustment* (ME1SCA3.PDF).

3-4 Dock Motor On Switch Removal

1. Refer to *Section 3-1-1* for Dock removal instructions.
2. Remove four mounting bolts from main dock cover. (See Illustration 3-5.)
3. Slide main dock cover forward until Up Limit Switch pushrod just clears main dock cover holes, then lift cover up. Unplug dock motor power lead before totally removing dock cover. (See Illustration 3-5.)
4. Remove front cover by removing two mounting nuts. (See Illustration 3-6.)
5. Remove and tag leads from TS1 and dock motor to Motor On Switch. (See Illustration 3-7.)



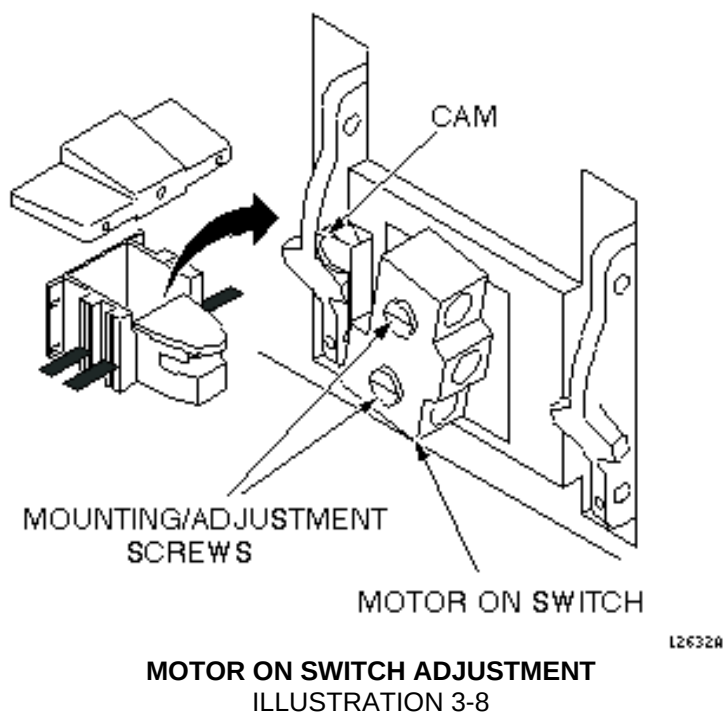
REPLACEMENT OF MOTOR ON SWITCH
ILLUSTRATION 3-7

L263-1A

6. Remove two mounting screws holding Motor On Switch to side of dock assembly.
7. Remove dock Motor On Switch.

3-5 Dock Motor On Switch Replacement

1. Secure new Motor On Switch to side of dock with two mounting screws.
 - a. Make sure that Motor On Switch actuates (clicks) when Table Up pedal is pressed approximately 3/4 through its length of travel. Adjust switch actuation by observing cam closure when pressing Table Up pedal. (See Illustration 3-7.)
 - b. Tighten two adjustment screws when proper setting is achieved. See Illustration 3-8.



2. Connect the leads to the new dock Motor On Switch as shown in Illustration 3-7.
3. Using an ohmmeter, check that there is continuity between black lead on dock motor power connection and blue lead at Pin 6 on TS1 when Table Up pedal is pressed approximately 3/4 through its length of travel.

4. Ensure that Motor On Switch operates properly by connecting an ohmmeter to Cable MG2 A29-P1, and performing a continuity check per Table 5.

TABLE 5
DOCK MOTOR ON SWITCH CONTINUITY CHECKS

Pins	Dock Up Pedal Position	Ohmmeter Reading
1, 11	Up	Open
2, 12	Down	~1 ohm
3, 13		
4, 23		
15, 24		
16, 25		
7, 20		

5. Replace front cover by securing two nuts. (See Illustration 3-6.)
6. Reconnect power lead to dock motor.
7. Carefully position main dock cover so that UP LIMIT switch push rod is aligned with upper hole in main dock cover. (See Illustration 3-5.)
8. Replace main dock cover by securing four mounting bolts. (See Illustration 3-5.)



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9. Refer to *Section 3-1-2* for Dock installation instructions.
10. Ensure that MOTOR ON switch operates properly when Table Up pedal is pressed.

3-7 Table Up Limit Switch Replacement

1. Install new table Up Limit Switch on dock bracket and secure with two mounting screws.
2. Attach push-on connectors to table Up Limit Switch.
3. Ensure that table Up Limit Switch operates properly by connecting an ohmmeter to Cable MG2 A31-J2, and performing a continuity check per Table 6.

TABLE 6
TABLE UP LIMIT SWITCH SPRING RETURN CHECK

Pins	Up Limit Switch Position	Ohmmeter Reading
9, 14	In	0 ohms
10, 15	Out	Open
1, 6	In	180 $\pm$ 10 ohms
2, 7	Out	180 $\pm$ 10 ohms

4. Verify that the minimum switch spring return force is 1.5 lbs. Press Up Limit pushrod, and take this reading with a force gauge on the Up Limit Switch pushrod at the point where the switch closes. (See Illustration 3-10.)

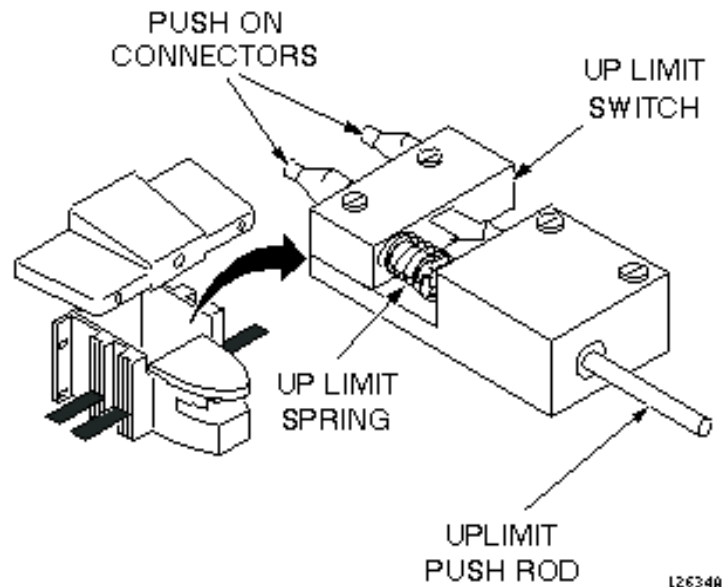


TABLE UP LIMIT SWITCH SPRING
ILLUSTRATION L2634A

5. Replace front cover by securing two mounting nuts. (See Illustration 3-6.)
6. Reconnect power lead to dock motor.
7. Carefully position main dock cover so that Up Limit Switch pushrod is aligned with upper hole in main dock cover. (See Illustration 3-5.)
8. Replace main dock cover by securing four mounting bolts. (See Illustration 3-5.)



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9. Refer to *Section 3-1-2* for Dock installation Instructions.
10. Check operation of the Up Limit Switch to ensure that it is activated when the table is within 1/4 inch (6.4 mm) of its full up position.

REVISION HISTORY

REV	DATE	AUTHOR	PRIMARY REASON FOR CHANGE
0	June 16, 1998	J. Saperstein	Initial conversion from Toolbook to Word.
1	May 21, 1999	S.M. Atladottir	Updated procedure references for new GUI.
2	Sept. 30, 2010	R. Kohl	Per CAPA 1859392: • Section 3 Dock Components - Added Warning. • Section 3-1 Dock Assembly - Added Dock Assembly Removal procedure, updated Dock Assembly Installation procedure, and added Warning. • Section 3-2 Dock Removal and Replacement - Updated Removal procedure and added Replacement procedure. • Section 3-4 Dock Motor On Switch Removal - Updated. • Section 3-6 Table Up Limit Switch Removal - Updated.